

Tema

A Machine Learning Approach to Predicting Firm Leverage in the UK Market: Which Features Matter Most, and How Does This Vary Across Firm Life Cycle Stages?

Research Questions

RQ1: Which firm-level features most accurately predict leverage in the UK market when using machine learning approaches?

RQ2: Does the relative importance of these predictors vary across firm life cycle stages, and if so, which features dominate at which stage?

Motivation

Capital structure literature identifies dozens of potential leverage determinants across trade-off, pecking order, and market timing theories. Existing ML applications (Amini et al. 2021) demonstrate that ML outperforms linear models but focus on US data. UK-specific ML analysis is largely absent despite institutional differences (LSE structure, UK bankruptcy law, tax regime, dividend culture) that may produce different feature rankings. Furthermore, feature importance may vary systematically with firm life cycle stage — a heterogeneity dimension not systematically examined in ML capital structure research.

Contribution

1. First systematic ML-based feature ranking for leverage prediction on UK market data (2005–2023)
 2. Novel life cycle heterogeneity analysis — how feature importance shifts across Introduction, Growth, Mature, Shakeout, Decline stages
 3. Practical output for practitioners: actionable feature ranking for UK credit analysis and firm valuation
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Data

- **Source:** Bloomberg Terminal (UCL SoM)
- **Universe:** UK non-financial non-utility publicly listed firms
- **Period:** 2005–2023 (annual data)
- **Filters:** Country of Incorporation = UK, exclude GICS 40 (Financials) and GICS 55 (Utilities), include Active + Delisted + Inactive
- **Size:** ~3000 securities → after cleaning ~1500–2000 unique firms × 19 years

Feature Set (Pooled Approach)

Target variable:

- Book Leverage = Total Debt / Total Assets

Firm-level features (~20):

- Tangibility, Firm size (log Assets), Altman Z-score, Effective tax rate, Non-debt tax shields
- Profitability, Cash holdings, Financing deficit, R&D intensity, Dividend dummy, Growth in assets
- Market-to-book, Cumulative stock returns, Stock return volatility, Baker-Wurgler efwamb
- Industry median leverage
- **Lagged leverage** (leverage_{t-1})
- **Operating cash flow / Assets**
- **Firm age** (from IPO date if available)

Life cycle classification:

- Dickinson (2011) — signs of Operating CF, Investing CF, Financing CF → 5 stages

Models

- **Baseline:** OLS regression
- **ML:** Lasso, Random Forest, XGBoost
- **Evaluation:** Out-of-sample R², RMSE, MAE
- **Train/test split:** Time-based (train 2005–2019, test 2020–2023)

Interpretation

- **SHAP values** on best-performing model for feature importance ranking
- **Global SHAP:** overall feature importance across all firms
- **Stage-specific SHAP:** feature importance within each life cycle stage
- **Comparison:** how does top-10 feature ranking shift between stages?

Analysis Structure

Part 1 (RQ1): Full sample analysis

- Train all models on full panel
- Compare R² across OLS, Lasso, RF, XGBoost

- SHAP ranking on best model

Part 2 (RQ2): Life cycle heterogeneity

- Classify each firm-year into Dickinson stage
 - Train models separately within each stage
 - Compare feature importance rankings across stages
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Dissertation Structure

1. **Introduction** — RQ, motivation, gap, contribution
 2. **Literature Review** — capital structure theories overview, ML in corporate finance, life cycle literature
 3. **Data & Methodology** — Bloomberg, sample, features, Dickinson classification, models
 4. **Results** — Part 1 (full sample), Part 2 (life cycle)
 5. **Discussion** — interpretation, comparison with US findings, practical implications
 6. **Conclusion** — contribution, limitations, future research
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План реализации (implementation roadmap)

Phase 1 — Data preparation (Week 1–2)

- Step 1.1** — Import Bloomberg Excel files into Python (pandas)
Step 1.2 — Merge separate variable sheets into single panel dataset (firm × year format)
Step 1.3 — Handle missing values (drop firms with <5 years of data)
Step 1.4 — Winsorize outliers at 1st and 99th percentile per year
Step 1.5 — Save cleaned panel as `panel_raw.csv`
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Phase 2 — Feature engineering (Week 2–3)

- Step 2.1** — Construct target variable: Book Leverage = Total Debt / Total Assets
Step 2.2 — Build all firm-level features (list of ~20 features from methodology)
Step 2.3 — Compute industry median leverage (group by GICS × year)
Step 2.4 — Compute lagged leverage (previous year for each firm)
Step 2.5 — Compute Baker-Wurgler efwamb (weighted historical M/B)
Step 2.6 — Apply Dickinson (2011) life cycle classification based on cash flow signs
Step 2.7 — Save final feature dataset as `panel_features.csv`
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Phase 3 — Exploratory Data Analysis (Week 3)

- Step 3.1** — Descriptive statistics for all features (mean, median, std)
 - Step 3.2** — Distribution of firms across life cycle stages
 - Step 3.3** — Leverage distribution by year, sector, life cycle stage
 - Step 3.4** — Correlation matrix of features
 - Step 3.5** — Missing data patterns visualization
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Phase 4 — Modeling (Week 4–6)

- Step 4.1** — Time-based train/test split (train 2005–2019, test 2020–2023)
 - Step 4.2** — Baseline: OLS regression, record R^2 , RMSE
 - Step 4.3** — Lasso regression with cross-validation for alpha
 - Step 4.4** — Random Forest with hyperparameter tuning (GridSearchCV or Optuna)
 - Step 4.5** — XGBoost with hyperparameter tuning
 - Step 4.6** — Save all models and evaluation metrics
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Phase 5 — Life cycle heterogeneity (Week 6–7)

- Step 5.1** — Segment training data by Dickinson stage (5 subsets)
 - Step 5.2** — Train separate models within each stage
 - Step 5.3** — Compare out-of-sample performance across stages
 - Step 5.4** — Save stage-specific models
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Phase 6 — SHAP interpretation (Week 7–8)

- Step 6.1** — Global SHAP values on best model (full sample)
 - Step 6.2** — Rank top-15 features by mean absolute SHAP
 - Step 6.3** — SHAP dependence plots for top 5 features
 - Step 6.4** — Stage-specific SHAP on stage-specific models
 - Step 6.5** — Comparison table: top-10 features per stage
 - Step 6.6** — Visualization: how ranking shifts across life cycle
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Phase 7 — Writing (Week 8–10)

- Step 7.1** — Draft Data & Methodology chapter (first — while methods fresh)
 - Step 7.2** — Draft Results chapter (with all tables and figures)
 - Step 7.3** — Draft Literature Review chapter
 - Step 7.4** — Draft Discussion chapter (compare findings with Amini US results)
 - Step 7.5** — Draft Introduction (last — знаешь что важно highlighted)
 - Step 7.6** — Draft Conclusion
 - Step 7.7** — Full revision, formatting, references
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Technical stack

Task	Tool
Data manipulation	pandas
ML models	scikit-learn, xgboost
Interpretation	shap library
Visualization	matplotlib, seaborn
Statistics	scipy, statsmodels
Environment	Jupyter notebooks + version control via git

Key deliverables per phase

Phase	Deliverable
1	<code>panel_raw.csv</code>
2	<code>panel_features.csv</code>
3	EDA notebook with visualizations
4	Trained models + metrics table
5	Stage-specific models + comparison table
6	SHAP plots + feature rankings
7	Final dissertation document